



OLLSCOIL NA hÉIREANN MÁ NUAD
THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

Year 3

Autumn Repeat Exam

2014 - 2015

Quantum Mechanics I
MP363

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Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. The Planck blackbody formula says that the spectral radiance per unit frequency emitted by a blackbody at temperature T is

$$B(\nu, T) = \frac{2h}{c^2} \frac{\nu^3}{e^{h\nu/k_B T} - 1}.$$

- (a) Rayleigh and Jeans computed the same quantity using only classical methods and obtained the Rayleigh-Jeans formula

$$B_{\text{RJ}}(\nu, T) = \frac{2k_B T \nu^2}{c^2}.$$

Show (by computing the Taylor series of $B(\nu, T)$ around $\nu = 0$ or any other method) that if $\nu \ll k_B T/h$, then

$$B(\nu, T) = B_{\text{RJ}}(\nu, T) + O(\nu^3),$$

and thus the two formulas approximately agree for low frequencies.

- (b) The energy flux per unit frequency emitted by a blackbody at temperature T is

$$\frac{d\Phi}{d\nu} = \pi B(\nu, T).$$

The Stefan-Boltzmann law says that the total energy flux from a blackbody due to radiation of all frequencies is

$$\Phi = \sigma T^4$$

where σ is a constant. Derive this relation and give the value of σ in terms of the fundamental constants h , c and k_B . (The integral

$$\int_0^\infty \frac{u^3}{e^u - 1} du = \frac{\pi^4}{15}$$

may be useful.)

2. Suppose A is an operator on a Hilbert space $\mathcal{H}$ with inner product $\langle | \rangle$ and $A^\dagger$ its adjoint.

- (a) Prove that, for any ψ and ζ in $\mathcal{H}$,

$$\langle \psi | A \zeta \rangle = \langle \zeta | A^\dagger \psi \rangle^*$$

- (b) The creation and annihilation operator of the quantum-mechanical simple harmonic oscillator act on wavefunctions as

$$\begin{aligned} a\psi(x) &= \sqrt{\frac{m\omega}{2\hbar}} \left[x\psi(x) + \frac{\hbar}{m\omega} \frac{\partial \psi}{\partial x}(x) \right], \\ a^\dagger \psi(x) &= \sqrt{\frac{m\omega}{2\hbar}} \left[x\psi(x) - \frac{\hbar}{m\omega} \frac{\partial \psi}{\partial x}(x) \right]. \end{aligned}$$

Use the inner product

$$\langle \psi | \zeta \rangle = \int_{-\infty}^{\infty} \psi^*(x) \zeta(x) dx$$

and the result from (a) to show that $a^\dagger$ is indeed the adjoint of a . State *explicitly* any assumptions you make about the behaviour of the functions ψ and ζ in this proof, and justify them.

3. The energy eigenfunctions for the infinite square well are $\phi_n(t, x) = u_n(x)e^{-iE_nt/\hbar}$ for $n = 1, 2, \dots$, where

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2, \quad u_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) & 0 \leq x \leq a, \\ 0 & x < 0, x > a \end{cases}$$

Suppose the state of a particle in such a well is given by the wavefunction

$$\psi(t, x) = \frac{1}{\sqrt{2}}\phi_1(t, x) - \frac{1}{\sqrt{2}}\phi_3(t, x).$$

- (a) If the energy of this state is measured, what are the possible results, and with what probabilities do they occur?
- (b) Compute $\langle X \rangle$ and $\langle P \rangle$ as functions of time.
- (c) Use your results from (b) to confirm that

$$\langle P \rangle = m \frac{d}{dt} \langle X \rangle.$$